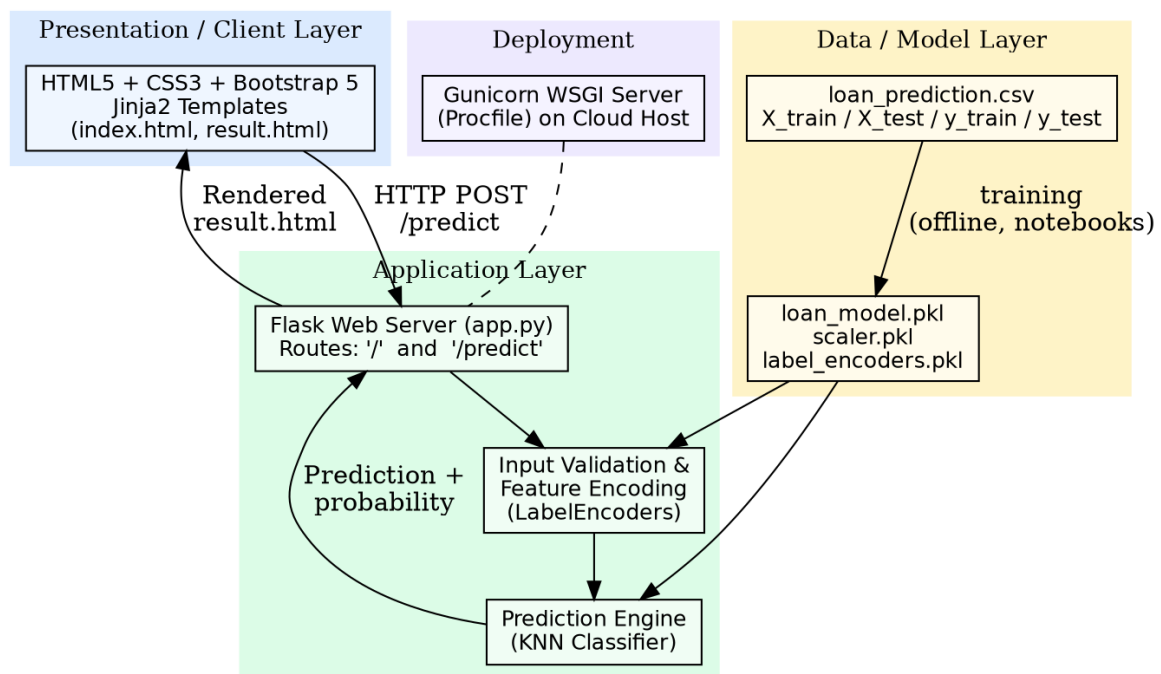


Solution Architecture

Date	03 July 2026
Team ID	XXXXXX
Project Name	Smart Lender - AI Powered Loan Eligibility Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

The diagram below shows the high-level architecture of the Smart Lender solution, structured into a Presentation Layer, Application Layer, Data/Model Layer, and Deployment Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	User interface for entering applicant data and viewing results	HTML5, CSS3, Bootstrap 5, Jinja2
Application Layer (Flask Server)	Receives form submissions, orchestrates validation and prediction	Python, Flask
Core Logic / Prediction Engine	Encodes and scales input, runs the trained model, computes confidence & risk	Scikit-learn (KNN)
Data / Model Layer	Stores the trained model, scaler, encoders and datasets	Joblib (.pkl), CSV
Deployment Layer	Serves the Flask app in production	Gunicorn, Procfile, cloud host